Advanced Data Analysis

These are the same subjects measured before (pre_score) and after (post_score) an intervention. Did their scores change significantly?

已停止思考 ›

Yes. Because the same subjects were measured before and after the intervention, the appropriate analysis is a **paired-samples t-test**.

Results

- Number of subjects: **25**
- Mean pre-score: **48.49**
- Mean post-score: **52.82**
- Mean change (post − pre): **+4.32**

The paired t-test showed a statistically significant increase in scores:

$$t(24) = 2.83, \quad p = 0.009$$

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Conclusion

The intervention was associated with a **significant improvement** in scores. Since $p < 0.05$, we reject the null hypothesis that the mean change is zero.